Unit 2.3 Limit laws (p 95)

Note Title 2014/02/24

1| Limit laws — p95 - 97

Limit Laws Suppose that c is a constant and the limits

$$\lim_{x \to a} f(x) \quad \text{and} \quad \lim_{x \to a} g(x)$$

exist. Then

1. $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$

2. $\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$

3. $\lim_{x \to a} [cf(x)] = c \lim_{x \to a} f(x)$

4. $\lim_{x \to a} [f(x)g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$

5. $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ if $\lim_{x \to a} g(x) \neq 0$

6. $\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x)\right]^n$ where n is a positive integer

7. $\lim_{x \to a} c = c$

8. $\lim_{x \to a} x = a$

9. $\lim_{x \to a} x^n = a^n$ where n is a positive integer

10. $\lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a}$ where n is a positive integer

    (If n is even, we assume that $a > 0$.)

11. $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$ where n is a positive integer

    [If n is even, we assume that $\lim_{x \to a} f(x) > 0$.]

Examples

1. $\lim_{x \to 2} \left(3x^2 + 4\sin\frac{\pi}{x} + 5\right)$

$= \lim_{x\to 2} 3x^2 + \lim_{x \to 2} 4\sin\frac{\pi}{x} + \lim_{x\to 2} 5$

$= 3\lim_{x\to 2} \underset{x \cdot x}{x^2} + 4\lim_{x\to 2} \sin\frac{\pi}{x} + 5$

$= 3\lim_{x\to 2} x \cdot \lim_{x\to 2} x + 4 \sin\frac{\pi}{\lim_{x\to 2} x} + 5$

$= 3(2)(2) + 4\sin\frac{\pi}{2} + 5$

$= 12 + 4 + 5 = 21$

2. $\lim_{x\to 3} \frac{x^2 + \sin x}{x} = \frac{9+\sin 3}{3}$ $\left(\frac{9+\sin 3}{3}\right)$

DIRECT SUBST. *

3. $\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3} \frac{\cancel{(x-3)}(x+3)}{\cancel{x-3}}$ $\left(\frac{0}{0}\right)$

Factorization *

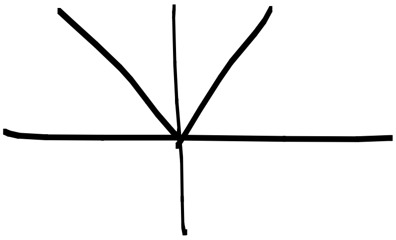
$= 6$

4. $\lim_{t\to 0} \frac{t}{\sqrt{t+4}-2} \times \frac{\sqrt{t+4}+2}{\sqrt{t+4}+2}$ $\left(\frac{0}{0}\right)$

$= \lim_{t\to 0} \frac{t(\sqrt{t+4}+2)}{(t+4)-4} = \lim_{t\to 0} \frac{\cancel{t}(\sqrt{t+4}+2)}{\cancel{t}}$

$= 4$

5. $\lim_{x \to 0} |x|$



$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

$= 0$

$\longrightarrow$

$\lim_{x \to 0^-} -x = 0$ $\qquad$ $\lim_{x \to 0^+} x = 0$

Theorem 1. p 99

$$\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = L = \lim_{x \to a^+} f(x)$$

Examples:

6. $\lim_{x \to 0} \dfrac{\sqrt{3+x} - \sqrt{3}}{x} \quad \left(\dfrac{0}{0}\right)$

$$= \lim_{x \to 0} \frac{\sqrt{3+x} - \sqrt{3}}{x} \times \frac{\sqrt{3\times x} + \sqrt{3}}{\sqrt{3+x} + \sqrt{3}}$$

$$= \lim_{x \to 0} \frac{(\cancel{3}+\cancel{x}) - \cancel{3}}{\cancel{x}(\sqrt{3+x} + \sqrt{3})} = \frac{1}{2\sqrt{3}}$$

7. $$\lim_{x \to 0} \frac{\frac{1}{3+x} - \frac{1}{3}}{x} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \to 0} \frac{\frac{3-(3+x)}{3(3+x)}}{x}$$

$$= \lim_{x \to 0} \frac{-\cancel{x}}{9+3x} \times \frac{1}{\cancel{x}} = -\frac{1}{9}$$

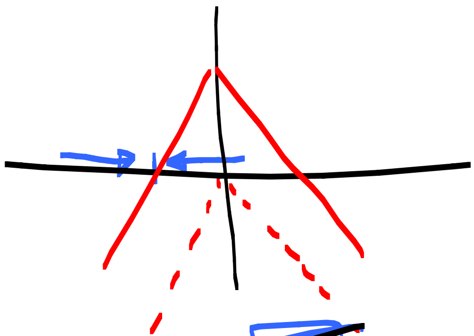
8. $$\lim_{x \to -2} \frac{2-|x|}{2+x} \quad \left(\frac{0}{0}\right)$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$= \lim_{x \to -2} \frac{2-(-x)}{2+x}$$

since the $x \to -2$ $\to$ $x < 0$

$$= \lim_{x \to -2} \frac{2+x}{2+x} = 1$$



9. $$\lim_{x \to 4} \frac{2-\sqrt{x}}{4x-x^2} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \to 4} \frac{2-\sqrt{x}}{x(4-x)} = \lim_{x \to 4} \frac{\cancel{2-\sqrt{x}}}{x\cancel{(2-\sqrt{x})}(2+\sqrt{x})}$$

$$= \frac{1}{16}$$

10. $$\lim_{x \to 0} \frac{|x|}{x} \quad \left(\frac{0}{0}\right) \qquad |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

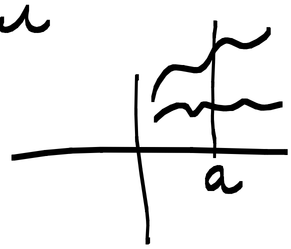
$$\lim_{x \to 0^-} \frac{-x}{x} = -1 \qquad \lim_{x \to 0^+} \frac{x}{x} = 1$$

$$\lim_{x \to 0} \frac{|x|}{x} \text{ DNE.}$$

Theorem 2. p101

If $f(x) \le g(x)$ when x is near a (except possibly at a), then

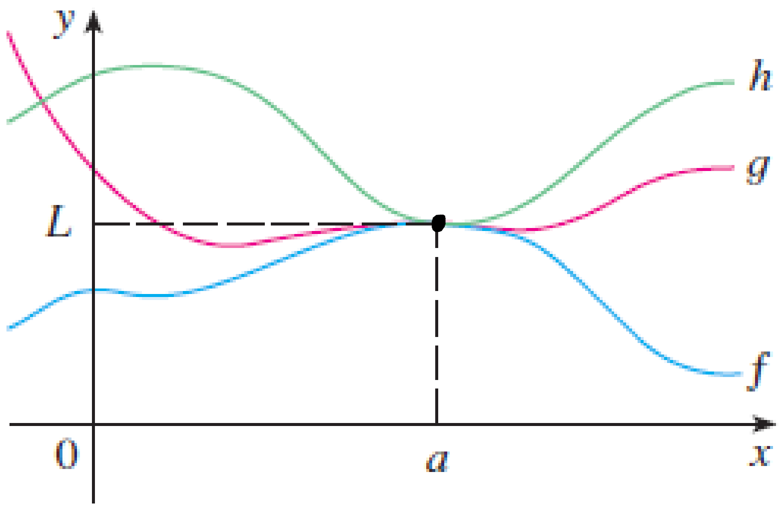
$$\lim_{x \to a} f(x) \le \lim_{x \to a} g(x)$$



Theorem 3. The squeeze theorem. p101

If $f(x) \le g(x) \le h(x)$ when x is near a and $\lim_{x \to a} f(x) = L = \lim_{x \to a} h(x)$ then

$$\lim_{x \to a} g(x) = L$$



Example: ① $\lim_{x \to 0} \sin\left(\frac{\pi}{x}\right)$ (See figure 7 p86)

DNE.

② $\lim_{x \to 0} x^2 \sin\left(\frac{\pi}{x}\right)$

$$-1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1$$

$$-x^2 \leq x^2 \sin\left(\frac{\pi}{x}\right) \leq x^2$$

$\lim_{x \to 0} -x^2 = 0$ and $\lim_{x \to 0} x^2 = 0$ $\Rightarrow$

$$\lim_{x \to 0} x^2 \sin\left(\frac{\pi}{x}\right) = 0$$

See figure 8 p 102

Look out for functions that are bounded such as:

$$-1 \leq \sin x \leq 1$$

$$-1 \leq \cos x \leq 1$$

$$-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$$

② $\lim_{x \to 0^+} \sqrt{x} \arctan x$ (Could use direct subst.)

$$-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$$

$$\Rightarrow \quad -\frac{\pi}{2}\sqrt{x} < \sqrt{x} \arctan x < \sqrt{x}\,\frac{\pi}{2}$$

$$\lim_{x \to 0^+} -\frac{\pi}{2}\sqrt{x} = \lim_{x \to 0^+} \sqrt{x}\,\frac{\pi}{2} = 0$$

$$\Rightarrow \lim_{x \to 0^+} \sqrt{x} \arctan x = 0$$

③ $\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$

$$-1 \leq \cos\left(\tfrac{1}{x}\right) \leq 1$$

$$\Rightarrow \quad -x^2 \leq x^2 \cos\left(\tfrac{1}{x}\right) \leq x^2$$

And $\lim_{x \to 0} (-x^2) = \lim_{x \to 0} x^2 = 0$

$\Rightarrow \lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right) = 0$ (Squeeze theorem)

Special limits in study guide p 14.

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$ [Example 3 p86 - no proof required] $\left(\frac{0}{0}\right)$

① $\lim_{x \to \pi} \frac{\sin x}{x} = 0$

② $\lim_{x \to 0} \frac{\tan x}{x} = \lim_{x \to 0} \frac{\frac{\sin x}{\cos x}}{x}$ (0)

$= \lim_{x \to 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x}$

$= \underbrace{\lim_{x \to 0} \frac{\sin x}{x}} \lim_{x \to 0} \frac{1}{\cos x}$

$= 1 \cdot 1 = 1$

③ $\lim_{x \to 0} \frac{x}{\sin x} = \lim_{x \to 0} \frac{1}{\frac{\sin x}{x}}$ $\left(\frac{0}{0}\right)$

$= \frac{1}{1} = 1$

④ $\lim_{x \to 0} \frac{x}{\tan x} = 1$

Accept.

$\lim_{x \to 0} \frac{\sin x}{x} = 1$

Examples:

① $\lim_{x \to 0} 4 \cdot \frac{\sin(4x)}{(4x)} = \lim_{x \to 0} 4 \lim_{x \to 0} \frac{\sin(4x)}{4x}$

$= 4 \cdot 1 = 4$

② $\lim_{x \to 2} \frac{\tan((x^2-4))}{((x-2)(x+2))} \times (x+2) = 1 \times 4 = 4$

③ $\lim_{x \to 0} \frac{x^2}{\sin x} = \underbrace{\lim_{x \to 0} \frac{x}{\sin x}} \cdot x = 1 \cdot 0$

$= 0$

④ $$\lim_{x \to 0} \frac{\tan(x^2)}{x} = \underbrace{\lim_{x \to 0} \frac{\tan(x^2)}{x \cdot x}} \times \frac{x}{1}$$

$$= 1 \times 0 = 0$$

⑤ $$\lim_{x \to 4} \frac{\sin(x-4)}{x^2-16} = \underbrace{\lim_{x \to 4} \frac{\sin(x-4)}{x-4}}_{1} \cdot \frac{1}{x+4}$$

$$= \frac{1}{8}$$

⑥ $$\lim_{x \to 4} \frac{\sin x}{x} = \frac{\sin 4}{4}$$

Squeeze theorem:

Find: $$\lim_{x \to 0^+} \sqrt{x}\, e^{\sin\left(\frac{\pi}{x}\right)}$$

We know that: $$-1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1$$

$$\Rightarrow e^{-1} \leq e^{\sin\left(\frac{\pi}{x}\right)} \leq e^{1}$$

$$\Rightarrow \frac{\sqrt{x}}{e} \leq \sqrt{x}\, e^{\sin\left(\frac{\pi}{x}\right)} \leq e\sqrt{x}$$

But $$\lim_{x \to 0^+} \frac{\sqrt{x}}{e} = \lim_{x \to 0^+} e\sqrt{x} = 0$$

$\Rightarrow$ By the squeeze theorem

$$\lim_{x \to 0^+} \sqrt{x}\, e^{\sin\left(\frac{\pi}{x}\right)} = 0.$$